SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I ____________________ (V. Asha Sarthiwika / 192311066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *V. Asha Sarthiwika*

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. AI-Powered Drones for Emergency Medicine Delivery (Greedy Best-First Search & A)*

i. Greedy Best-First Search (GBFS): Behavior, Strengths and Weaknesses

How it works

- Greedy Best-First Search selects the path that appears closest to the destination.
- It uses only the heuristic value h(n) (straight-line distance).
- It ignores the actual travel cost.

Strengths

- Finds a route very quickly.
- Uses less memory.
- Suitable when fast decisions are required.
- Easy to implement.
- Works well if the heuristic is accurate.

Weaknesses

- May choose blocked or risky paths.
- Does not guarantee the shortest path.
- Can get stuck due to incorrect heuristic.
- Ignores terrain and weather costs.
- Performs poorly in highly dynamic environments.

*ii. How A Search Works**

A* considers both:

- g(n) = Actual cost travelled
- h(n) = Estimated remaining distance

Formula:

f(n) = g(n) + h(n)

Working

- Calculates total estimated cost.
- Chooses the node with the smallest f(n).
- Updates the path whenever better information is available.
- Avoids expensive or dangerous routes.

Advantages

- Finds the optimal path.

- Considers both distance and travel cost.
- Handles variable terrain costs better.
- Safer than Greedy Search.
- More reliable in emergencies.

iii. Impact of Heuristic Accuracy

For Greedy Search

- Accurate heuristic → Fast and good solution.
- Poor heuristic → Wrong path and delays.

For A*

- Accurate heuristic → Faster search.
- Less accurate heuristic → More nodes explored.
- Still finds the optimal solution (if heuristic is admissible).

iv. Effect of Dynamic Changes (Blocked Paths)

- Floods may suddenly block roads.
- Weather may increase travel cost.
- Both algorithms must replan routes.

Greedy Search

- Easily misled by blocked paths.
- May require many repeated searches.

A*

- Updates costs and searches another path.
- Better handles changing conditions.
- More reliable under uncertainty.

v. Best Algorithm Recommendation

Recommended: A*

Reason

- Produces optimal routes.
- Considers actual travel cost.
- Handles dynamic updates effectively.
- Provides safer delivery.
- Balances speed and accuracy.

Conclusion: A* is the best choice because emergency medicine delivery requires safety, minimum delay, and reliable path planning.

2. Smart City Traffic Signal Optimization (Hill Climbing & Simulated Annealing / Genetic Algorithm)

Optimization Problem

Objective

Minimize:

- Total waiting time
- Fuel consumption
- Traffic congestion

Why Traditional Search is Inefficient

- Huge number of signal combinations.
- Traffic changes every minute.
- Searching every possibility takes too much time.
- Computational cost is very high.
- Not suitable for real-time systems.

i. Hill Climbing

Working

- Starts with an initial signal timing.
- Makes small changes.
- Chooses the better neighbouring solution.
- Stops when no better solution is found.

Advantages

- Fast.
- Easy to implement.
- Uses little memory.
- Good for simple optimization.
- Suitable for quick improvements.

Limitations

- Gets trapped in local maxima.
- Cannot escape plateaus.
- May stop before finding the best solution.

- Depends on starting point.
- Poor for complex traffic systems.

ii. Local Maxima, Plateaus and Ridges
Local Maximum
A good solution that is not the best.
Example: AI reduces congestion in one area but ignores other intersections.

Plateau
Many neighbouring solutions have the same value.
Example: Changing signal timing gives almost no improvement.

Ridge
Best solution requires several coordinated changes.
Example: One signal change alone is not enough; multiple intersections must change together.

iii. Simulated Annealing / Genetic Algorithm
Simulated Annealing
- Sometimes accepts worse solutions.
- Escapes local maxima.
- Finds better global solutions.
- Gradually reduces randomness.

Genetic Algorithm
- Maintains many candidate solutions.
- Uses selection, crossover and mutation.
- Searches many areas simultaneously.
- Produces high-quality solutions.
- Works well for large optimization problems.

iv. Best Recommendation
Recommended: Genetic Algorithm
Reason

- Highly scalable.
- Adapts to changing traffic.
- Avoids local optima.
- Optimizes multiple objectives together.
- Suitable for large smart city systems.

3. Mars Rover Exploration (Online Search Agent)
i. Why Online Search?
Offline search assumes:
- Complete map.
- Known environment.

Mars rover:
- Environment is unknown.
- Discovers obstacles while moving.
- Learns continuously.

Therefore, Online Search is required.

ii. Challenges
Partial Observability
- Cannot see the entire terrain.
- Limited sensors.
- Hidden rocks and craters.

Dynamic Environment
- Dust storms.
- Loose soil.
- New obstacles.
- Sensor uncertainty.

These make planning difficult.

iii. Internal Model
The rover continuously:
- Collects sensor data.
- Stores explored locations.
- Marks hazards.
- Updates safe routes.

- Improves future decisions.

Its map becomes more accurate over time.

iv. Online Search vs Other Searches

Feature	Uninformed	Informed	Online
Map Required	Yes	Yes	No
Uses Heuristic	No	Yes	Yes (when available)
Unknown Environment	Poor	Limited	Excellent
Adaptability	Low	Medium	High

v. Best Recommendation

Recommended: Online Search Agent

Reason

- Works without complete knowledge.
- Learns while exploring.
- Adapts to new hazards.
- Makes real-time decisions.
- Ensures safer navigation.

4. University Exam Timetable (Constraint Satisfaction Problem)

Problem Formulation

Variables

- Exam time for each course.
- Exam hall assignment.
- Faculty assignment.

Domains

Possible:

- Time slots.
- Exam halls.
- Available faculty.

Constraints

1. No student has overlapping exams.

2. Hall capacity must not exceed limits.
3. Some exams occur before others.
4. Faculty must be available.
5. Students needing accommodations receive suitable arrangements.

ii. Backtracking Search
Working

- Assign one exam at a time.
- Check constraints.
- If conflict occurs:
    - Undo assignment.
    - Try another option.
- Continue until all exams are scheduled.

Advantages

- Guarantees a valid solution if one exists.
- Systematically explores possibilities.

iii. Constraint Propagation (Forward Checking)
Working

- Removes impossible choices early.
- Prevents future conflicts.
- Reduces unnecessary searching.

Benefits

- Faster search.
- Fewer backtracking steps.
- Better efficiency.
- Saves computation time.
- Suitable for large timetables.

iv. Real-World Challenges

- Thousands of students.
- Hundreds of courses.
- Limited halls.
- Faculty changes.

- Last-minute updates.

Best Strategy

Use:

- Backtracking
- Forward Checking
- Constraint Propagation

This combination improves scalability and reduces computation.

5. Strategic Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

Working

- AI assumes the opponent also plays optimally.
- AI (MAX) tries to maximize its score.
- Opponent (MIN) tries to minimize AI's score.
- AI selects the move with the best guaranteed outcome.

ii. Computational Challenges

- Huge game tree.
- Millions of possible moves.
- High memory usage.
- Slow computation.
- Difficult to search completely in real time.

iii. Alpha-Beta Pruning

Working

- Removes branches that cannot affect the final decision.
- Produces the same result as Minimax.
- Searches fewer nodes.

Advantages

- Faster decision making.
- Reduces computation.
- Searches deeper.
- Uses less memory.
- Suitable for real-time games.

iv. Evaluation Function
The evaluation function estimates the quality of a game state.
Factors

- Player resources.
- Army strength.
- Territory control.
- Unit health.
- Distance to objectives.
- Defensive position.
- Enemy strength.
- Available actions.

These factors help the AI choose the strongest move.

v. Best Recommendation
Recommended Strategy
Use:

- Minimax + Alpha-Beta Pruning + Evaluation Function

Reason

- Makes intelligent decisions.
- Greatly reduces search time.
- Handles large game trees efficiently.
- Maintains near-optimal performance.
- Suitable for real-time strategy games.